Deep Divs

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April 3, 2026

How to factorize

- **Completing the square:** $x^2+ax = \left(x+\frac{a}{2}\right)^2 - \frac{a^2}{4}$. You can then look at quadratic residues, or you can use this as part of a larger factoring. This can also set up a Pell's equation[2] (e.g. $x^2 + x = 2y^2$).

- **Difference and sums of nth powers:**

$$x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \ldots + xy^{n-2} + y^{n-1}),$$
$$x^n + y^n = (x+y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 - \ldots - xy^{n-2} + y^{n-1}) \text{ if } n \text{ is odd.}$$

- **Sophie Germain's identity:** $x^4 + 4y^4 = (x^2 - 2xy + 2y^2)(x^2 + 2xy + 2y^2)$.

- **Polynomials evaluated at two different points:** There are many problems that revolve around the identity $x - y \mid P(x) - P(y)$ for P a polynomial with integer coefficients.

- **Diophantine equations with variables in the exponents** (e.g. $3^x - 2^y = 1$). For these problems, you almost always want to use modular arithmetic to show a couple exponents have to be multiples of some integer n, and then factor the equation as a difference (or sum) of n^{th} powers.

The Art of Choosing a Mod

- Make n a prime power. The Chinese Remainder Theorem guarantees that looking at any polynomial mod xy is no better than looking at it mod x and then looking at it mod y (assuming x and y are relatively prime). If there are variables in the exponents, you might want to break this rule, but even then, prime powers are still usually the right choice.

- If there are perfect squares in the equations, try n a power of 2. The fewer quadratic residues there are mod n, the better off you will be. If n is a power of 2, the number of quadratic residues mod n is $\left[\frac{n}{2}\right] + 1$. If n is a power of

$p \neq 2$, the number of quadratic residues mod n is $\left\lfloor \frac{pn}{2(p+1)} \right\rfloor \geq \left[\frac{n}{3}\right]$. So, as you can see, powers of 2 are basically twice as good as the other choices! In practice, 4, 8, and sometimes 16 are good numbers to try.

- If there are m^{th} powers in the equation, the key is to choose $n = p^k$ so that $g = \gcd(m, (p-1)p^{k-1})$ is as large as possible. This is because the number of m^{th} powers mod n is approximately $\frac{n}{g}$. Usually you want to choose p, k so that $m \mid (p-1)p^{k-1}$.
- Make sure there is something to gain from doing modular arithmetic in the first place! The technique is very useful for showing an equation has no solutions. But if it has even one solution, it will also have a solution mod n for all n. Even if you can show all solutions are 1 mod 1,000,000,000, that still leaves an infinite number of possibilities to check!
- If the sum of the digits of an integer $S(n)$ is involved, always consider mod 9, because $S(n) \equiv n \pmod 9$.

Problem 1

Find all pairs of positive integers (a, b) such that

$$19^{19} = a^3 + b^4$$

Problem 2

Find all prime numbers p and q such that

$$p^3 - 3^q = 10.$$

Problem 3

Find all pairs of positive integers (a, b) such that

$$3^x - 2^y = 7$$

Problem 4

Find all positive integers n such that $2^n + 2021n$ is a perfect square.

Problem 5

Find all positive integers (a, b) such that $a^2 + 4b$ and $b^2 + 4a$ are both perfect squares.

Problem 6

Find all pairs of positive integers (a, b) such that $a > b$ and

$$7^a - 3^b \mid a^4 + b^2$$

Problem 7

Given a sequence of positive integers $a_1, a_2, \ldots, a_n$, where $n \geq 3$. Prove that there exists two indices i, j such that $a_i + a_j \nmid x$, where x can be any element from the set $\{3a_1, 3a_2, \ldots, 3a_n\}$.

Problem 8

Let $a_1, a_2, \ldots, a_n$ be a permutation of the set $\{1, 2, \ldots, n\}$. where $n \mid a_i(a_{i+1} - 1)$, where $i \neq k$. Prove that $n \nmid a_n(a_1 - 1)$